# SimpleFT8 — Two Antennas, One Comparison

What does Diversity actually deliver? — 40m FT8, first field results

Period: 2026-04-20 to 2026-05-11 · 15s cycles evaluated: 34986

## Summary:


With Diversity Standard (blue) I decoded on average +59% more stations than with a single antenna.
Including 'rescued' stations, this rises to up to +77%.
Diversity DX (orange) delivers +34% — less than Standard, but targeted at weak DX signals.
Basis: average over all measurement points from 4 days, all hours of the day — true daily average.


*Note on antenna setup: ANT1 is a Kelemen DP-201510 (trap dipole for 20m/15m/10m) — operated off-band on 40m and therefore suboptimal on this fre*

## What do the three modes mean?

Normal (grey): One antenna, no special logic — classic single-antenna setup. This is the reference baseline.
Diversity Standard (blue): Two antennas. The system automatically picks the antenna with more stations.
Diversity DX (orange): Two antennas. Picks the antenna with the weakest DX signals (below −10 dB).
Rescue (green caps): Stations ANT1 couldn't hear — ANT2 decoded them anyway.

# How was it measured?

Setup, time period and a bit of background

## The Setup

Measured on 40m FT8 using a FlexRadio FLEX-8400M — two antenna ports, same frequency.
Period: 2026-04-20 to 2026-05-11. Data available across the full day (05–23 UTC), night measurements ongoing.
Cycles: Normal 10844 (5 day/s) | Diversity Standard 13649 (6 day/s) | Diversity DX 10493 (5 day/s).
Each FT8 cycle lasts 15 seconds — the app counts decoded stations per cycle.

## Why not just use the daily average?

Good question — I did it that way at first too. But if one day had only 20 cycles and another had 500, the short day would carry the same weight. That's unfair. So all individual values are pooled and averaged directly — regardless of which day. This gives a picture much closer to reality.

## What are 'rescued stations' (Rescue)?

Imagine a station transmitting so weakly that ANT1 receives it below −24 dB.
That is the FT8 decoding threshold — below that, decoding is practically impossible.
ANT2 receives the same station with slightly more signal and decodes it anyway.
We call this 'Rescue'. The green caps in the charts show exactly these stations.
Whether you count them or treat them separately — that's up to you. Both values are in this report.

## Where does the raw data come from?

SimpleFT8 writes a small Markdown file per hour containing all cycle values.
No pre-computed averages — raw data only. The analysis script calculates everything fresh.
Want to check? statistics/ in the GitHub repo, all public.

# Main Results — Comparison Table

40m FT8 · Daily average across 4 measurement days, all hours of day

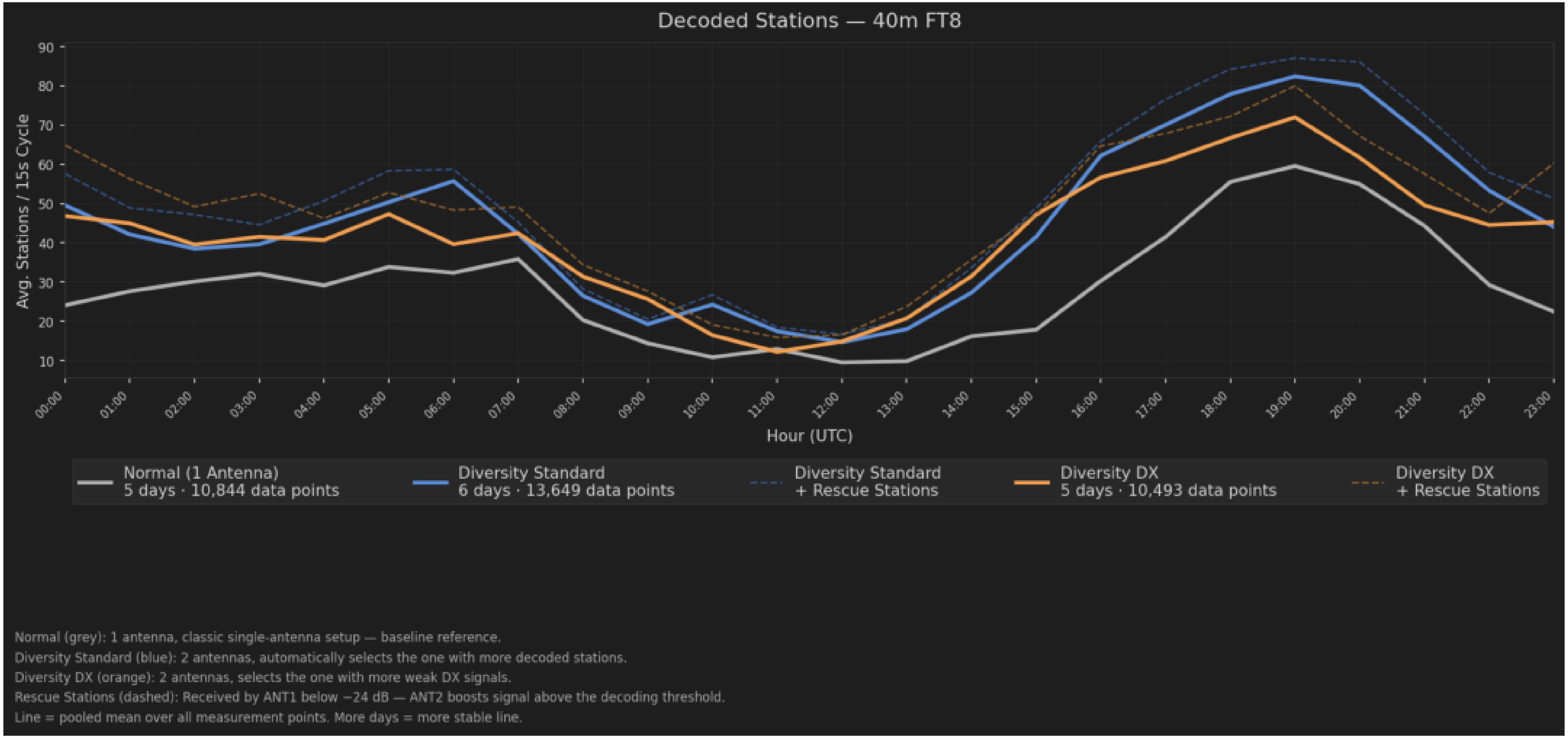
| Mode | Avg Sta. /15s Cycle | vs Normal w/o Rescue | vs Normal + Rescue | Rescue only | Meas. Days | Cycles |
|---|---|---|---|---|---|---|
| Normal (1 Antenna | 30.5 | — | — | — | 5 | 10844 |
| Diversity Standard | 48.5 | **+59%** | **+77%** | **+18%** | 6 | 13649 |
| Diversity DX | 40.8 | **+34%** | **+56%** | **+22%** | 5 | 10493 |

Avg Sta./15s Cycle: How many stations were decoded simultaneously per 15-second FT8 cycle on average — averaged over all measurement points (cycles) from 4 measuremen

*Note: ANT1 (Kelemen DP-201510) is operated off-band on 40m (design band: 20m/15m/10m) — ANT2 (house gutter ~15m) happens to be better suited for 40m. Gains represent*

# Stations per Hour — all three modes compared

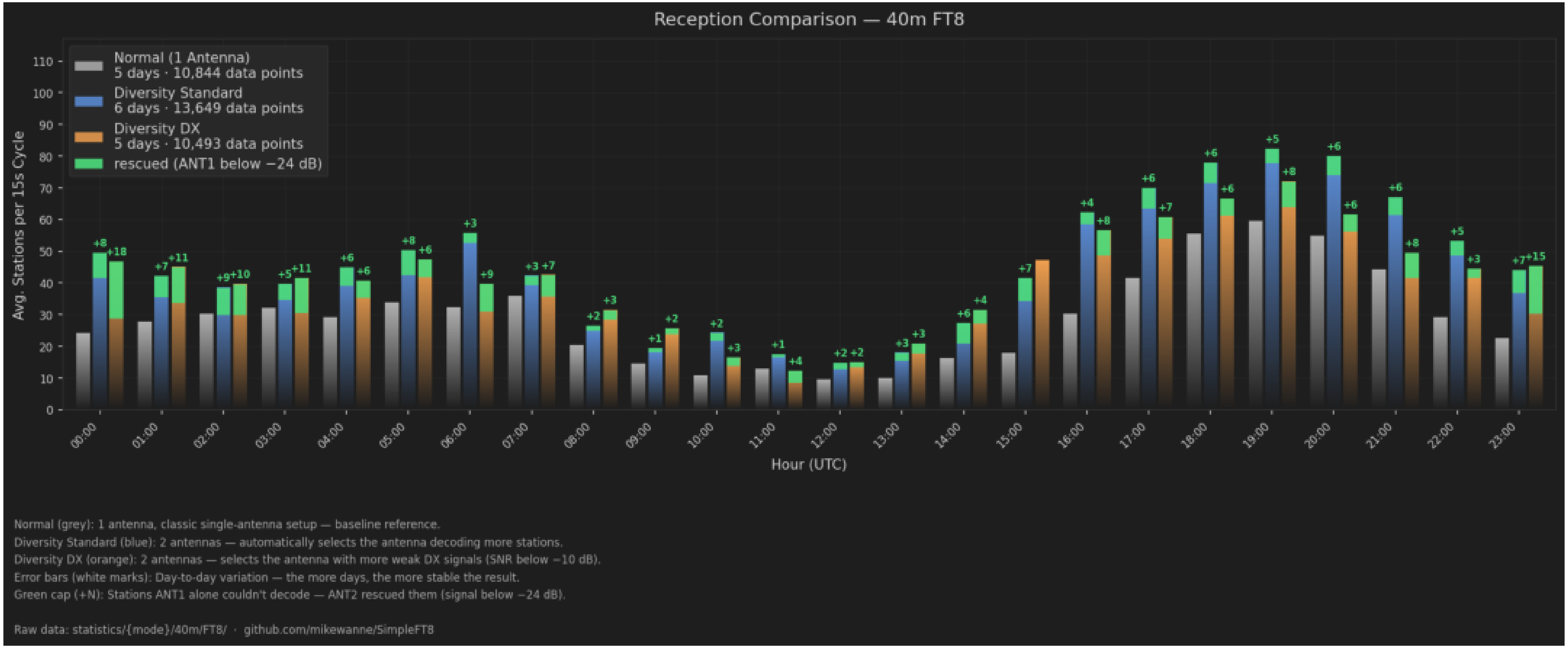
40m FT8 — line = pooled mean over all measurement days. Dashed = + rescue stations.




*You can clearly see how the grey line (Normal, one antenna) almost always falls below blue and orange. The dashed lines show the additional contribution of ANT2 rescue stations.*

# Direct Comparison — bars per hour, three modes side by side

40m FT8 — Normal (grey) | Diversity Standard (blue) | Diversity DX (orange) | Rescue caps (green)




*Green caps (+N) on top of bars show stations that ANT1 couldn't decode — signal below −24 dB. ANT2 rescued them anyway. Whether to count them or not — your call. White error bars show 33%–67% tertiles of slot values.*

# The Green Caps — count them or not?

Rescue stations: What are they and what do they tell us?

## What's this about?

In the diversity chart you can see small green caps with a +N on top of some bars.
These are stations ANT1 couldn't decode — their signal was below −24 dB, too weak.
ANT2 still received and decoded them. Over the measurement period this averaged
Ø 5.6 stations per hour for Diversity Standard, and Ø 6.8/h for Diversity DX.

## Why should you count them in?

Because the QSO is real for the operator — regardless of whether ANT1 or ANT2 made it possible.
These stations would never have made it into the log with a single antenna.
This is not a measurement artifact — it's exactly the point of running a second antenna.
With Rescue: Standard +77%, DX +56% — that's the full picture.

## Why might you leave them out?

SNR values fluctuate within the 15 seconds of a cycle — a station just
below −24 dB might be above that threshold in the next cycle.
The −24 dB threshold is an empirical value, not a law of nature.
For a clean comparison with other systems, counting only what ANT1 directly decoded is more conservative.

*That's why this report always shows both numbers: without Rescue (the conservative value) and with Rescue (what you get in real operation). Ta*

# What can you take away from this?

Conclusions and what gets measured next

## What is clearly visible:

Diversity Standard consistently delivers between 59% and 77% more stations across all measurement days — depending on whether you count rescued stations or not. This is not coincidence, it repeats.
Diversity DX sits at 34% without Rescue — less than Standard, but DX deliberately optimises for the weakest signals. If you do a lot of DX, that makes sense.
The strongest effect was at 15:00 UTC with +132% — typically the time when the band is just opening or closing.

## What cannot be said yet:

Whether results transfer 1:1 to other antenna combinations — probably not.
ANT1 operates off-band on 40m, ANT2 happens to be better suited there.
This explains the particularly large gain — with two matched 40m antennas it would be smaller.
Contest operation, geomagnetic storms, poor conditions — not yet systematically tested.
Whether this works identically on other transceivers — tested only on the FLEX so far.

## What comes next:

Night measurements on 40m are ongoing — 40m is often significantly better in the evening.
More days so the bars in the chart stabilize and variation decreases.
20m tests are planned — there ANT1 operates within its design band (20m/15m/10m) and the 20m band is generally easier to receive. Different
This report updates automatically whenever new data comes in.

*Raw data: statistics/ in the repo · github.com/mikewanne/SimpleFT8 · DA1MHH*